

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B.Tech (CBCS) Degree Examinations

Summer Semester July 2024

EE3002-1 – LINEAR CONTROL SYSTEMS
(EE)

Duration: 3 Hours

Max. Marks: 100

Note:

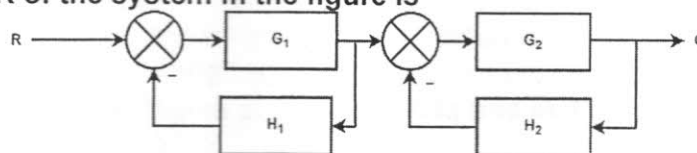
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- In force-voltage analogy, velocity is analogous to
A) Current B) Charge
C) Inductance D) Capacitance
- In a system, zero initial condition means that
A) The system is at rest and no energy is stored in any of its components B) The system is working with energy stored in the elements.
C) The system is working with zero reference signal D) any of the above
- Which of the following signal will become zero when the feedback signal and reference signs are equal.
A) Input B) Feedback
C) Reference D) Actuating
- In force-voltage analogy, Mass is analogous to
A) Charge B) Current
C) Inductance D) Resistance
- The transfer function C/R of the system in the figure is



- $$\frac{G_1 G_2}{1 + G_1 H_1 + G_2 H_2}$$
 - $$\frac{G_1 H_1 G_2 H_2}{(1 + G_1 H_1)(1 + G_2 H_2)}$$
 - $$\frac{G_1 G_2}{1 - G_1 - G_2 + G_1 G_2 H_1 H_2}$$
 - $$\frac{G_1 H_1 G_2 H_2}{1 + G_1 H_1 + G_2 H_2 + G_1 G_2 H_1 H_2}$$
- Which of the following transfer function of second order linear time-invariant systems represents an underdamped system?
A) $\frac{C(s)}{R(s)} = \frac{4}{s^2 + 4s + 4}$
B) $\frac{C(s)}{R(s)} = \frac{4}{s^2 + 5s + 4}$
C) $\frac{C(s)}{R(s)} = \frac{4}{s^2 + 4.5s + 4}$
D) $\frac{C(s)}{R(s)} = \frac{4}{s^2 + 3s + 4}$

7. The steady-state error due to unit-step input to a type-1 system is
- A) $\frac{1}{1+K_p}$ B) Zero
- C) ∞ D) $\frac{1}{K_p}$
8. Which one of the following is not the property of root loci?
- A) The root locus is symmetrical about imaginary axis B) They start from the open loop poles and terminate at the open loop zeroes
- C) The breakaway points are determined from $dK/ds = 0$ D) Segments of the real axis are the part of the root locus if and only is the total number of real poles and zeroes to their right is odd.
9. The characteristic equation of a feedback control system is given by $s^3 + 6s^2 + 9s + 4 = 0$. The number of roots in the left-half of s-plane is
- A) 3 B) 2
- C) 1 D) zero
10. A unity feedback system has forward transfer function $G(s) = \frac{K}{s(s+3)}$. The range of K for the system to be stable is
- A) $0 < K < 390$ B) $0 < K < 39$
- C) $0 < K < 3900$ D) $K > 0$
11. Consider a LTI system whose input $r(t)$ and $y(t)$ are related by the following differential equation:
- $$\frac{d^2y}{dt^2} + 4y = 6r(t)$$
- The poles of this system are at
- A) $+2j, -2j$ B) $+4j, -4j$
- C) $+2, -2$ D) $+4, -4$
12. Determine number asymptotes for $G(s)H(s) = \frac{(s+1)(s+4)}{(s+3)(s+5)}$
- A) Zero B) ∞
- C) 2 D) 4
13. If the constant 'K' is positive, then what would be its contribution on the phase plot?
- A) 0° B) 90°
- C) 180° D) -90°
14. At which frequency does the magnitude of the system becomes zero dB?
- A) Resonant frequency B) Cut-off frequency
- C) Gain crossover frequency D) Phase crossover frequency
15. The frequency at which the phase of the system acquires _____ is known as 'Phase crossover frequency'.
- A) 180° B) -180°
- C) -90° D) 90°
16. When the gain margin is positive and phase margin is negative, then the system is
- A) Unstable B) Stable
- C) Oscillatory D) Highly stable
17. The minimum number of states require to describe the two-degree differential equation:
- A) 1 B) 2
- C) 3 D) 4
18. In an RLC circuit, the number of state variables are
- A) 1 B) 0
- C) 3 D) 2
19. The input of a controller is
- A) Output signal B) Error signal
- C) Desired variable value D) None of the mentioned

20. Lag compensation leads to:

- A) Increased bandwidth
 B) Increases damping factor
 C) Attenuation
 D) None of the mentioned

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

1. a) List out the comparison between open-loop and closed-loop control systems
 b) Derive the expression for steady-state error for the step and ramp input signals.
 c) Using Mason's formula, find the transfer function $Y(s)/X(s)$ for the system whose signal flow graph is shown in Figure Q1(c)

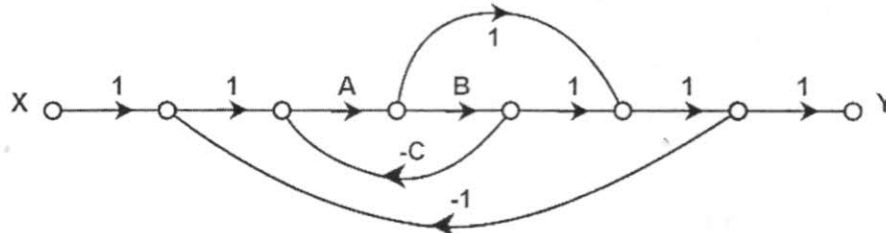


Figure Q1(c)

2. a) Determine the transfer function for the system shown in Fig Q2(a)

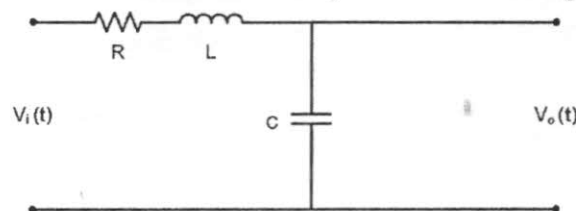


Figure Q2(a)

- b) Consider a second order system whose transfer function is given by

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

where $\xi = 0.6$ and $\omega_n = 5$ rad/sec. Calculate the rise time, peak time, maximum overshoot and settling time for both 2% and 5% criteria when the system is subjected to unit-step input.

3. a) Derive an expression for $c(t)$ of a first-order system for a unit step input.
 b) With the free body diagram of system, Write the differential equations governing the behavior of the mechanical translational system shown in Fig Q3(b) and obtain the force-voltage analogous network.

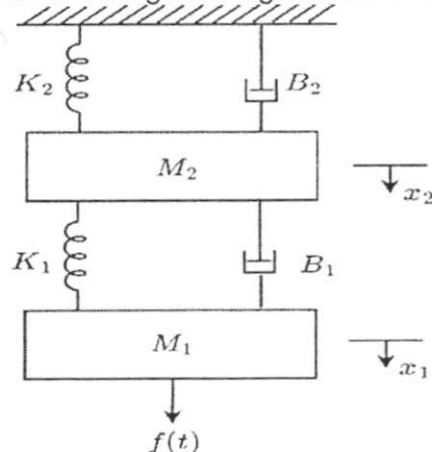


Figure Q3(b)

Marks	BT*	CO*	PO*
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04	L*2	1	1,2
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08	L2	2	1,2
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04	L3	1	1,2
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08	L3	1	1,2
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08	L3	2	1,2
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08	L3	2	1,2
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08	L3	1	1,2
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Unit – II

4. a) Examine the stability of the system whose characteristic equation is given by $S^5 + 2S^4 + 4S^3 + 8S^2 + 3S + 1 = 0$ using Routh-Hurwitz stability criterion 06 L3 3 1,2
- b) Draw the Bode magnitude and phase plot of the following open-loop transfer function $G(s)H(s) = \frac{1}{s(s+2)(s+4)}$ 10 L3 4 1,2
5. a) Write a note on the construction rules of root locus 08 L2 3 1,2
- b) Explain the step-by-step procedure obtaining the magnitude plot using Bode Plot. 08 L2 4 1,2
6. a) Sketch the root locus for the unity feedback system whose open loop transfer function is given by
- $$G(s)H(s) = \frac{K}{s(s+1)(s+2)}$$
- b) Sketch the magnitude plot for the system having $G(s) = \frac{5(1+2s)}{(1+4s)(1+0.25s)}$ using Bode Plot. 08 L3 3 1,2
- 08 L3 4 1,2

Unit – III

7. a) Write a note on lag, lead, and lag-lead compensators. 08 L2 5 1,2
- b) Construct a state model for a system characterized by the differential equation, $\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y + u = 0$. Also draw the block diagram representation of the state model. 08 L3 5 1,2
8. a) Write a note on the effect of P, PI and PD controllers 08 L2 5 1,2
- b) Obtain the state model of the system whose transfer function is given by $\frac{Y(s)}{U(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$ 08 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
